

Database Keys

Superkey, Candidate Key, Primary Key, and Foreign Key

Introduction

In relational databases, **keys** are fundamental concepts that help uniquely identify records (tuples) in tables (relations) and establish relationships between tables. This document explains four important types of keys:

- **Superkey**
- **Candidate Key**
- **Primary Key**
- **Foreign Key**

Each key plays a specific role in ensuring data integrity and avoiding redundancy.

Superkey

Definition

A **Superkey** is a set of one or more attributes (columns) that, taken collectively, can uniquely identify a tuple (row) in a relation (table).

Important Notes:

- It may include extra attributes that are not necessary for uniqueness.
- Every relation has at least one superkey — the set of all attributes.

Example of Superkeys

Consider a **Student** relation with the following attributes:

StudentID	Name	Email	Phone
101	Alice	alice@example.com	1234567890
102	Bob	bob@example.com	0987654321
103	Charlie	charlie@example.com	1112223333

Possible superkeys include:

- {StudentID}
- {StudentID, Name}
- {StudentID, Email, Phone}
- {Email, Phone} (*if combination uniquely identifies students*)

Each of these sets uniquely identifies every student.

Candidate Key

Definition

A **Candidate Key** is a *minimal* superkey — meaning it is a superkey with no redundant attribute. It uniquely identifies tuples, and no subset of it can be a superkey.

Important Notes:

- Candidate keys are minimal — they have no unnecessary attributes.
- There can be multiple candidate keys in a relation.
- Each candidate key can serve as a unique identifier.

Example of Candidate Keys

From the **Student** relation above:

- {StudentID} is a candidate key (minimal and unique).
- {Email} can be a candidate key if emails are unique.
- {StudentID, Name} is **not** a candidate key (not minimal).

Only {StudentID} and possibly {Email} qualify as candidate keys.

Primary Key

Definition

A **Primary Key** is the candidate key selected by the database designer to uniquely identify tuples in a relation.

Important Notes:

- Only one primary key can exist in a table.
- Primary key attributes cannot contain NULL values.
- It is used as the main reference for identifying records.

Example of Primary Key

In the **Student** relation, **StudentID** is usually chosen as the primary key because:

- It uniquely identifies each student.
- It does not contain null values.

Hence,

Primary Key = {StudentID}

Foreign Key

Definition

A **Foreign Key** is an attribute or set of attributes in one relation that refers to the *primary key* in another relation. It establishes a relationship between two tables and enforces referential integrity.

Important Notes:

- The foreign key value must match an existing primary key value in the referenced table or be NULL.
- It is used to represent relationships between tables (e.g., one-to-many).

Example of Foreign Key

Consider two relations:

- `Student(StudentID, Name, Email)`
- `Enrollment(EnrollmentID, StudentID, CourseID)`

Here,

`Enrollment.StudentID` $\rightarrow$ references $\rightarrow$ `Student.StudentID`

`StudentID` in `Enrollment` is a foreign key pointing to the `Student` table's primary key. This ensures every enrollment record refers to a valid student.

Summary Table

Key Type	Definition	Example
Superkey	Any set of attributes that uniquely identifies tuples (may contain extra attributes).	<code>{StudentID}</code> , <code>{StudentID, Name}</code> , <code>{Email, Phone}</code> (if unique)
Candidate Key	Minimal superkey, no redundant attribute.	<code>{StudentID}</code> , <code>{Email}</code> (if unique)
Primary Key	Selected candidate key used as the unique identifier for the table.	<code>{StudentID}</code>
Foreign Key	Attribute(s) referencing primary key of another table to establish relationships.	<code>Enrollment.StudentID</code> referencing <code>Student.StudentID</code>